MDP Project

Project Report

Multi-Pollutant Air-Quality Machine Learning over 204 Indian Monitoring Stations

Build -> Audit -> Verify -> Reframe. A progress report on a cloud-scale air-quality ML pipeline, its independent scientific audit, and the honest reframing of its contribution.

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Multi-Pollutant Air-Quality Machine Learning over 204 Indian Monitoring Stations *Build -> Audit -> Verify -> Reframe. Progress report.*

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1. Executive summary

The MDP project set out to build a cloud-scale machine-learning pipeline that goes from *prediction* of air-pollutant levels, to *causal structure* among pollutants, to *policy optimization* -- "Causal AI for Air Pollution." We **built the full pipeline, ran it on real data, and then subjected it to a rigorous independent audit**. The engineering is sound and reproducible, and the dataset is real; but the audit found that the strongest scientific claims (causal mechanisms, source apportionment, Pareto-optimal policy) are **not supported** by the available data.

We therefore **reframed the work honestly** as an *open, leakage-safe benchmark* for multi-pollutant air-quality ML -- a dataset, an evaluation protocol, and honest baselines others can build on -- plus a set of cautionary methodological findings. Measurement units were verified against the authoritative source, and all citations were verified. The project is, for review purposes, **complete**; what remains are decisions (venue, authorship) and a final reformat before submission.

2. Original objective

A three-phase PySpark pipeline on Google Cloud Dataproc:

- **Phase 1** -- species clustering (K-means) to group co-varying pollutants.
- **Phase 2** -- predictive baseline (XGBoost) + causal discovery (DirectLiNGAM).
- **Phase 3** -- multi-objective policy optimization (Optuna/NSGA-II) + explainability (SHAP).

Intended framing: *"Causal Machine Learning for Multi-Pollutant Source Apportionment and Control Optimization."*

3. What we built

- A complete **PySpark pipeline** (`src/phases/`, driven by `main.py`) implementing all three phases, packaged for **Google Cloud Dataproc** with a custom Docker runtime (`Dockerfile`, `submit.sh`, `cloudbuild.yaml`).
- An automated **data-collection tool** (`data_collection.py`) pulling real measurements from the OpenAQ v3 API and co-located weather from Open-Meteo.
- The pipeline **runs end-to-end** on the real 204-station dataset -- every phase completes successfully on Dataproc and in Cloud Shell.
- Full **engineering hygiene**: MIT licence, README, requirements, fixed seeds, reproducible manifests, and a correctness test suite.

(Tool-by-tool justification is in `SOFTWARE_STACK.md`; algorithm pseudocode in `PSEUDOCODE.md`.)

4. The dataset

- **204 Indian monitoring stations**, hourly, calendar year 2025.
- Pollutants: CO, NO, NO2, O3, PM10, PM2.5, SO2; merged with modelled weather (temperature, humidity, wind, pressure).
- **~1.18 million station-hours** (~66% of a full 204 x 8,760 grid); 37 stations below 50% within-year coverage.
- Real, open, and reproducible -- committed alongside the code.

5. The scientific audit

Acting as a critical reviewer, we independently re-examined every major claim on a dedicated branch (`review/scientific-audit`), writing self-contained scripts that regenerate every number offline with a fixed seed. The audit added the diagnostics the original pipeline lacked: **leakage-safe evaluation**, bootstrap stability, meteorology-confounding tests, and an exact-optimality cross-check.

6. Key findings

6.1 Prediction. Random-split evaluation gave an optimistic R2 ~ 0.41, but that is **data leakage** on autocorrelated data. Under honest, leakage-safe evaluation the held-out R2 is **0.25** (blocked-temporal) to **0.30** (unseen stations). Pollutants add only **+0.12 R2 beyond weather**; a 1-hour-ahead forecast beats a persistence baseline by only **+0.05**. The model beats trivial spatial baselines but not temporal persistence.

6.2 Causal structure. Of all discovered edges, exactly **one** is both bootstrap-stable and survives the inclusion of meteorology: **NO -> ozone (negative)**, consistent with local titration chemistry -- reported as a *hypothesis*, not an identified effect. The wider graph is confounded by weather.

6.3 Grouping. K-means produces only weak clusters (silhouette 0.26-0.39); it is *grouping*, not source apportionment (which needs speciation, uncertainties and VOCs the data lack).

6.4 Optimization. The published metaheuristic returned **19 "optimal" policies -- all of which are dominated** by the exact closed-form frontier we derived. A worker bug (wrong sampler) and a phantom decision variable were also found and fixed.

6.5 Units (major finding). Verified against the **authoritative OpenAQ sensor metadata**: NO (ppb), ozone, PM2.5, PM10 (µg/m3) are unit-consistent, but **CO, NO2 and SO2 carry duplicate ppb + µg/m3 sensors at almost every station** with no per-row provenance, so their stored unit is indeterminate. Crucially, both headline results (ozone prediction, NO->ozone) rest only on **unit-consistent** variables, so they stand.

7. The honest reframe

Because the words *causal*, *source apportionment* and *control optimization* could not be supported, we narrowed the claim rather than conceal the limitation. The work is now presented as:

> ***An open, leakage-safe benchmark for multi-pollutant air-quality machine learning*
>
> *over 204 Indian monitoring stations** -- contributing an open dataset, a*
>
> *leakage-safe evaluation protocol, honest baselines, one testable NO-O3 hypothesis,*
>
> *and a clear account of what stronger claims would require.*

This is a genuine, defensible contribution to the air-quality / environmental-AI field (see `SIGNIFICANCE.md`).

8. Deliverables produced

- Full pipeline + deployment (code, Docker, scripts)
- Real 204-station dataset
- Audit scripts reproducing every result (`audit/*.py`)
- Rough research-paper draft (`ROUGH_DRAFT.md`), honest and narrowed
- Pseudocode, software justification, significance, publication plan
- **6 publication figures** (regenerable; appended here)
- **All 12 references verified** against authoritative records
- Units verification against authoritative metadata
- Secondary-dataset evaluation (a modelled dataset, assessed and set aside)

- A shareable PDF package of all of the above

9. Current status & remaining work

Status: complete for review purposes. All *work* is done, committed and pushed.

Remaining items are decisions/formalities, not engineering:

1. Confirm the benchmark framing and choose a target venue (with the supervisor).
2. Finalize author list and affiliations.
3. Reformat the draft into the chosen venue's template; final proofread; submit.

10. One-paragraph conclusion

We built a working, reproducible, cloud-scale air-quality ML pipeline on real Indian data, then held it to a high evidentiary standard. Rather than overstate what the data support, we corrected course: a verified dataset, a leakage-safe evaluation, one stable scientific hypothesis, and an honest map of the limits -- released as an open benchmark. The result is a smaller but **defensible** contribution, and a project that demonstrates not just building a pipeline, but knowing what it does and does not prove.

Appendix - Figures

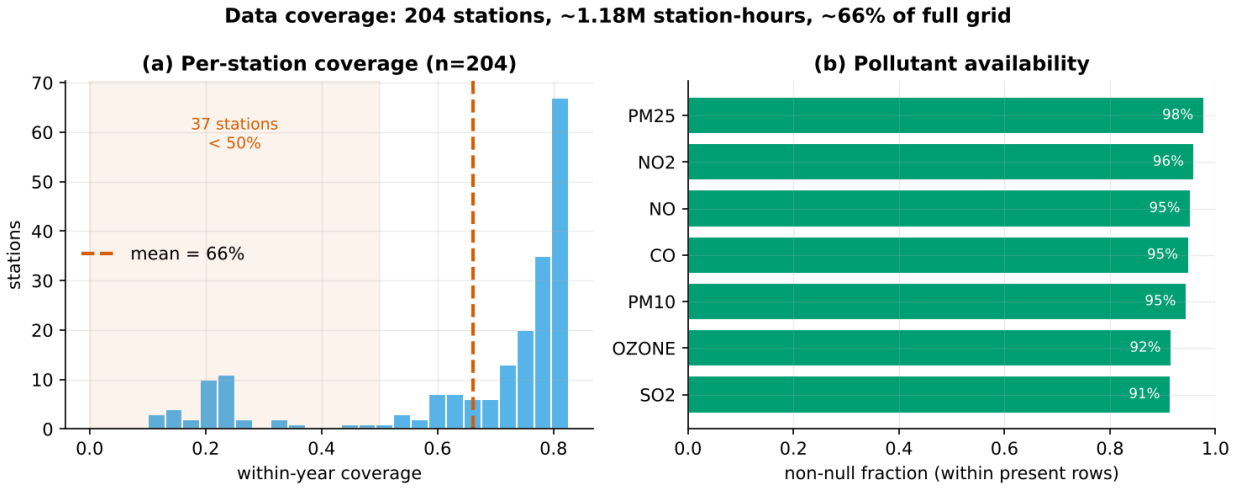


Fig. 1 - Data coverage: 204 stations, ~66% of the full grid; pollutant availability.

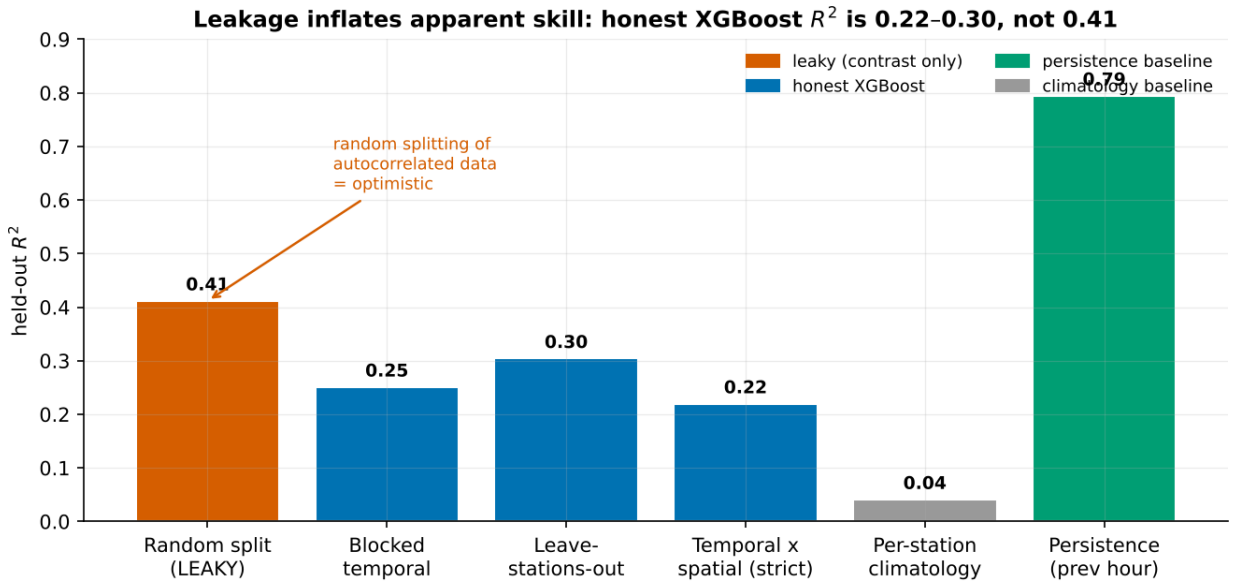


Fig. 2 - Leakage inflates skill: honest held-out R2 is 0.22-0.30, not 0.41.

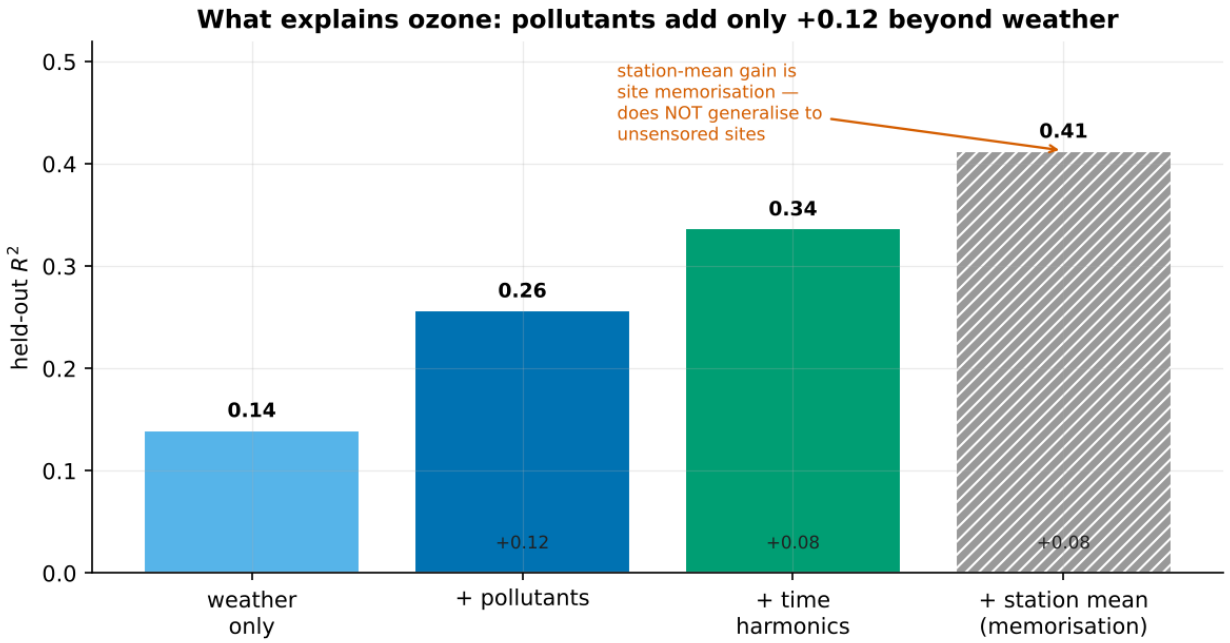


Fig. 3 - Pollutants add only +0.12 R2 beyond weather; station-mean gain is memorisation.

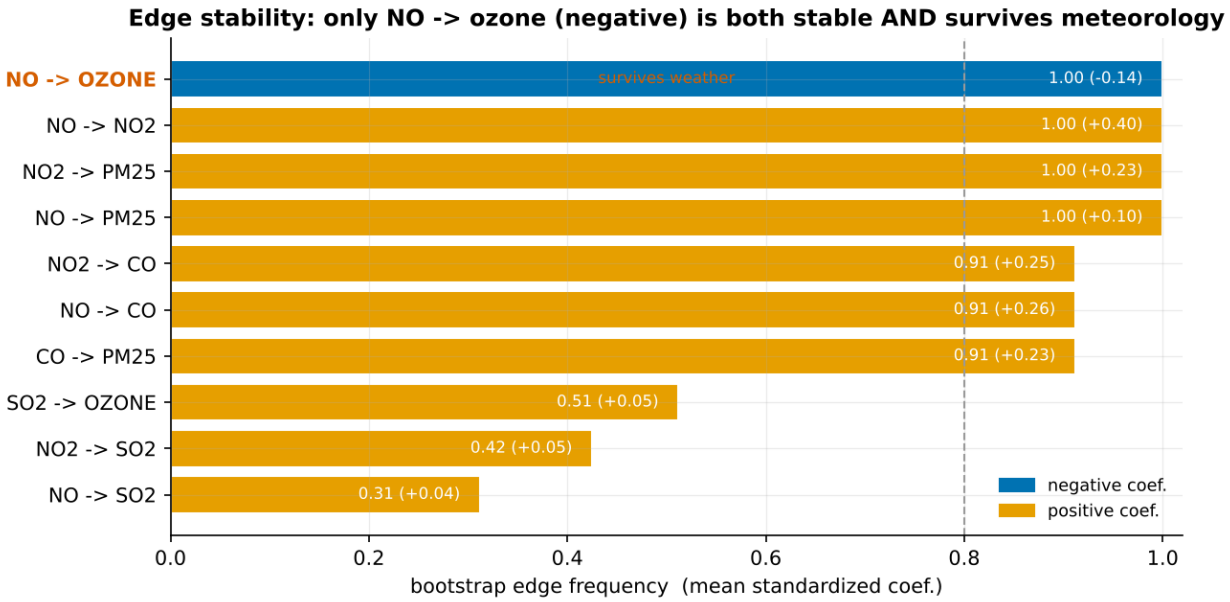


Fig. 4 - Edge stability: only NO -> ozone (negative) is stable AND survives meteorology.

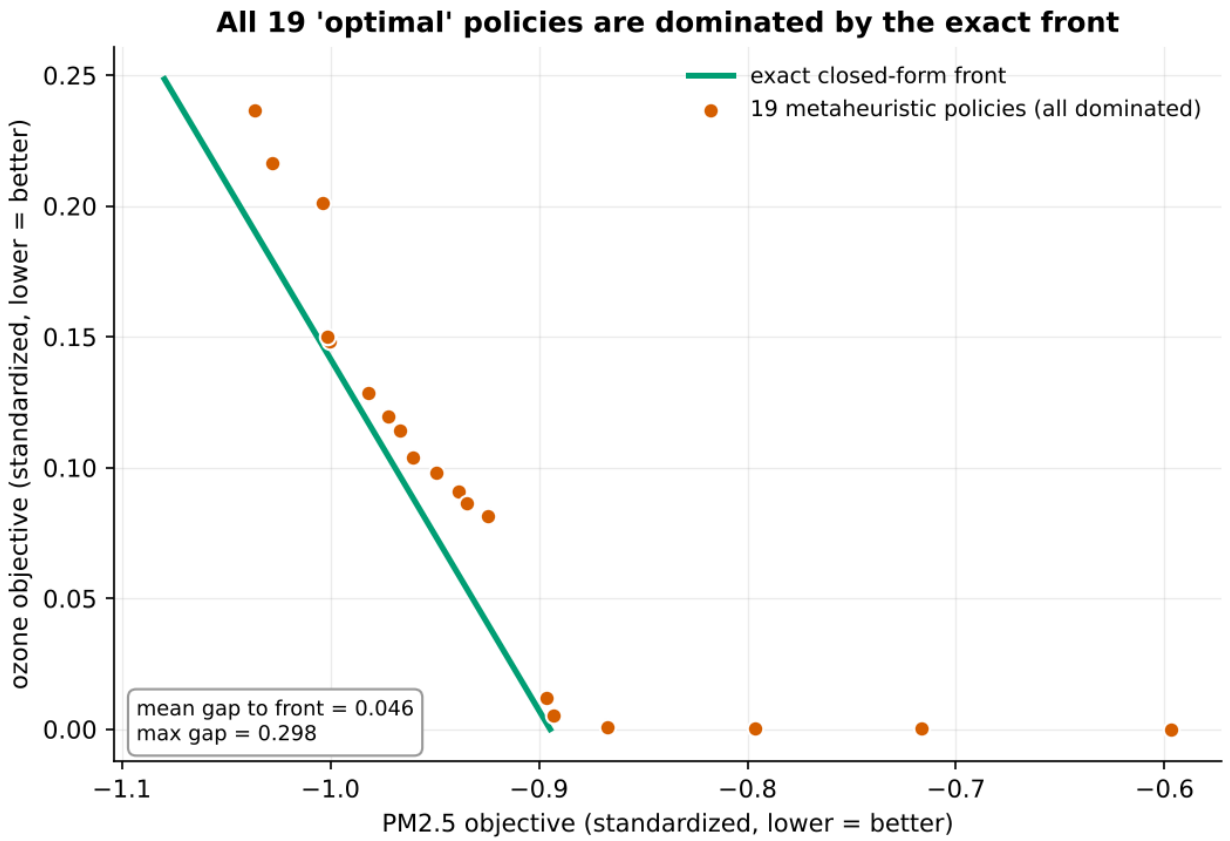


Fig. 5 - All 19 metaheuristic 'optimal' policies are dominated by the exact front.

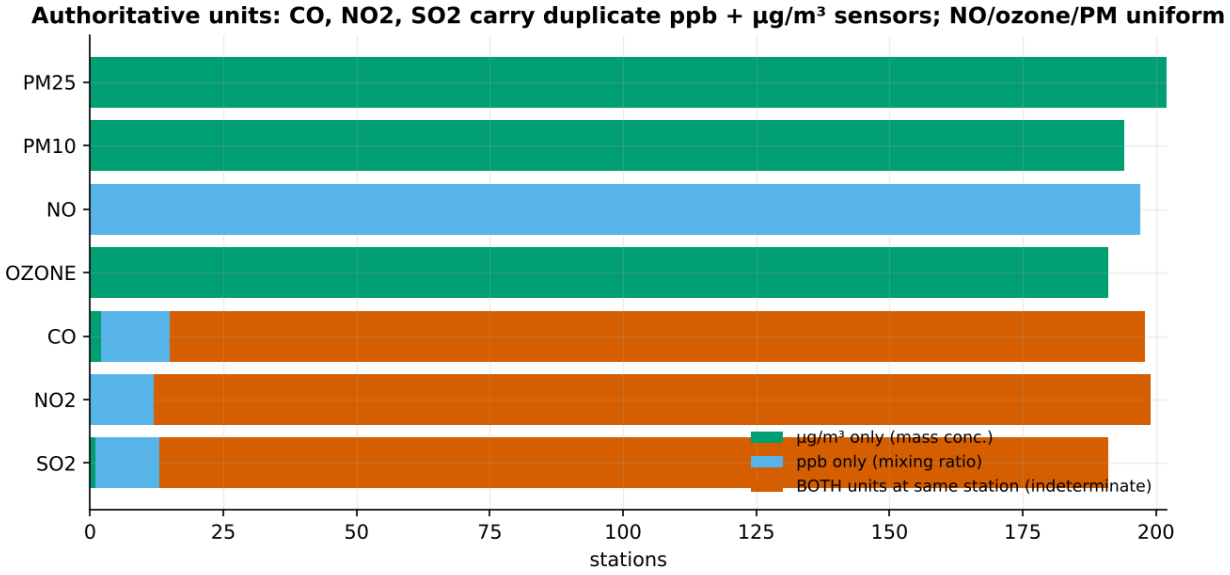


Fig. 6 - Authoritative units: CO, NO2, SO2 carry duplicate ppb + ug/m3 sensors (indeterminate).